

Retail Financial Performance & Forecasting Analytics

End-to-End Business Intelligence Project using SQL, Python & Power BI

Business Context

Retail companies operate across multiple product categories, regions and customer segments, making it difficult for leadership to maintain clear visibility into financial performance. Decision makers require consolidated insights into revenue trends, profitability drivers operational efficiency and future sales outlook.

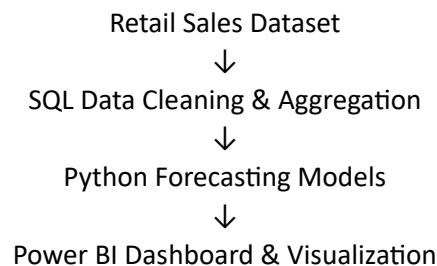
Technology Stack

SQL (MySQL) – Data cleaning and aggregation of transactional sales data

Python (Pandas, Statsmodels) – Time-series forecasting using SARIMA and Holt-Winters

Power BI (DAX, Data Modeling) – Interactive dashboards for financial performance analysis

Analytics Workflow



Data Analysis & SQL Processing

SQL was used to transform raw transactional data into structured metrics required for analysis. Aggregated outputs such as monthly revenue and profit trends formed the foundation for both dashboard reporting and forecasting models.

Example SQL Query

```
SELECT  
MONTH(order_date) AS month,  
SUM(revenue) AS total_revenue,  
SUM(profit) AS total_profit  
FROM fact_sales  
GROUP BY month;
```

These aggregated metrics were used to evaluate revenue growth, profitability patterns and sales performance across time periods.

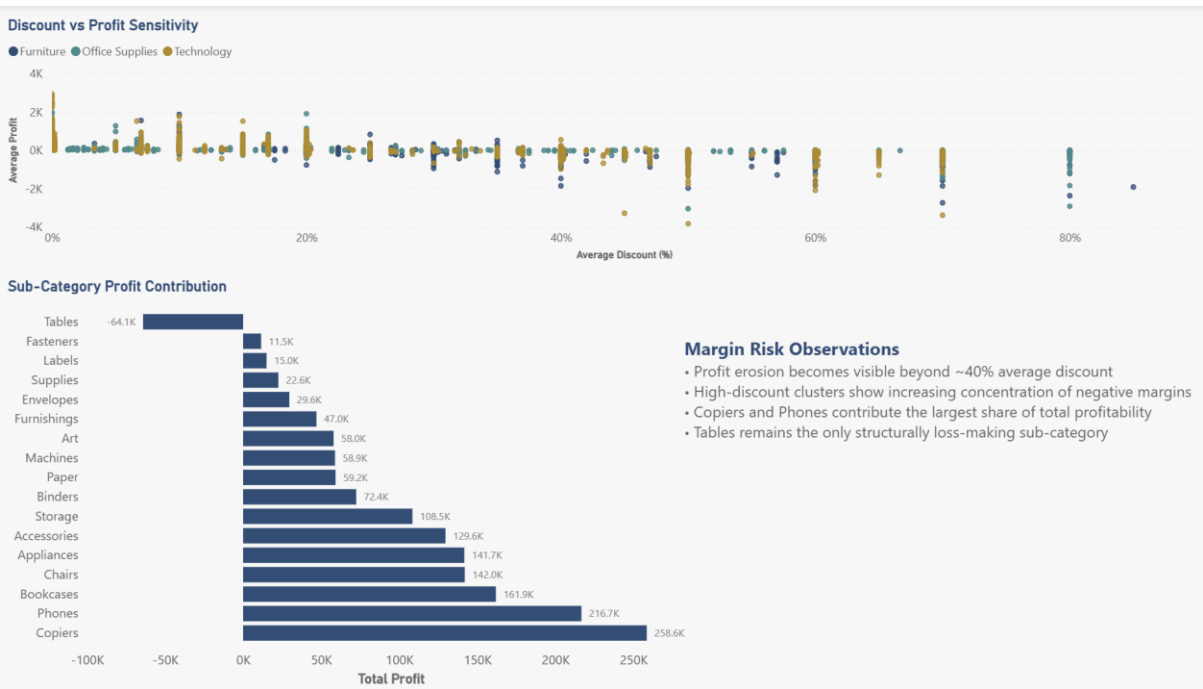
Data Insights

The Power BI dashboard analyzes retail financial performance across revenue trends, profitability, customer segments and operational metrics.



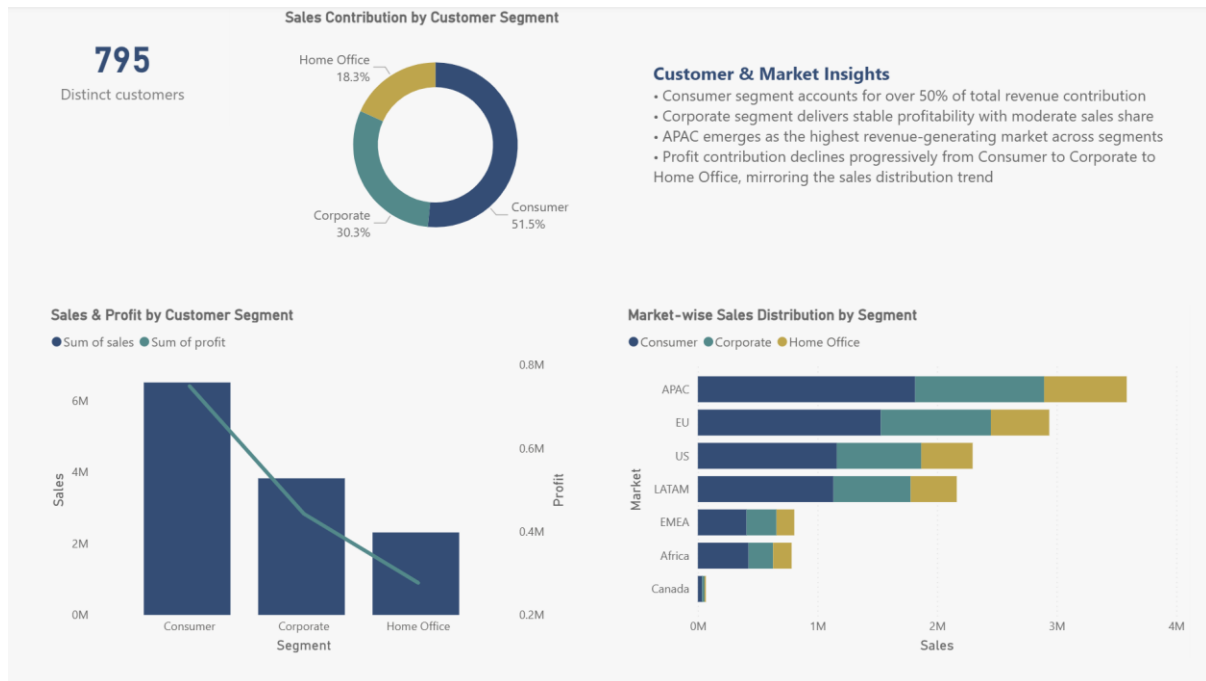
Executive Performance Summary

The analysis shows consistent revenue growth across the dataset with total revenue reaching **\$13M** and an average profit margin of **~11–12%**. Seasonal sales patterns are visible across the time series.



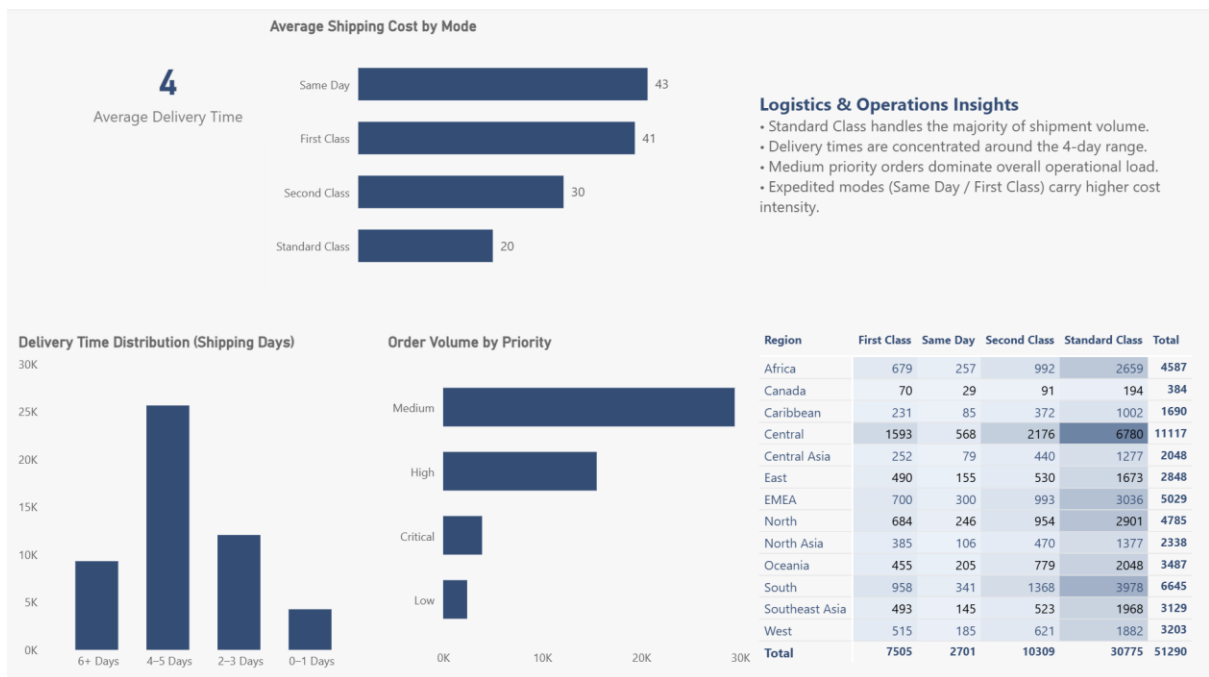
Margin & Discount Sensitivity

Profitability declines significantly when discount levels exceed **40%**. Certain product categories such as **Copiers and Phones** contribute strongly to overall profits, while **Tables** consistently show weaker margins.



Customer & Market Dynamics

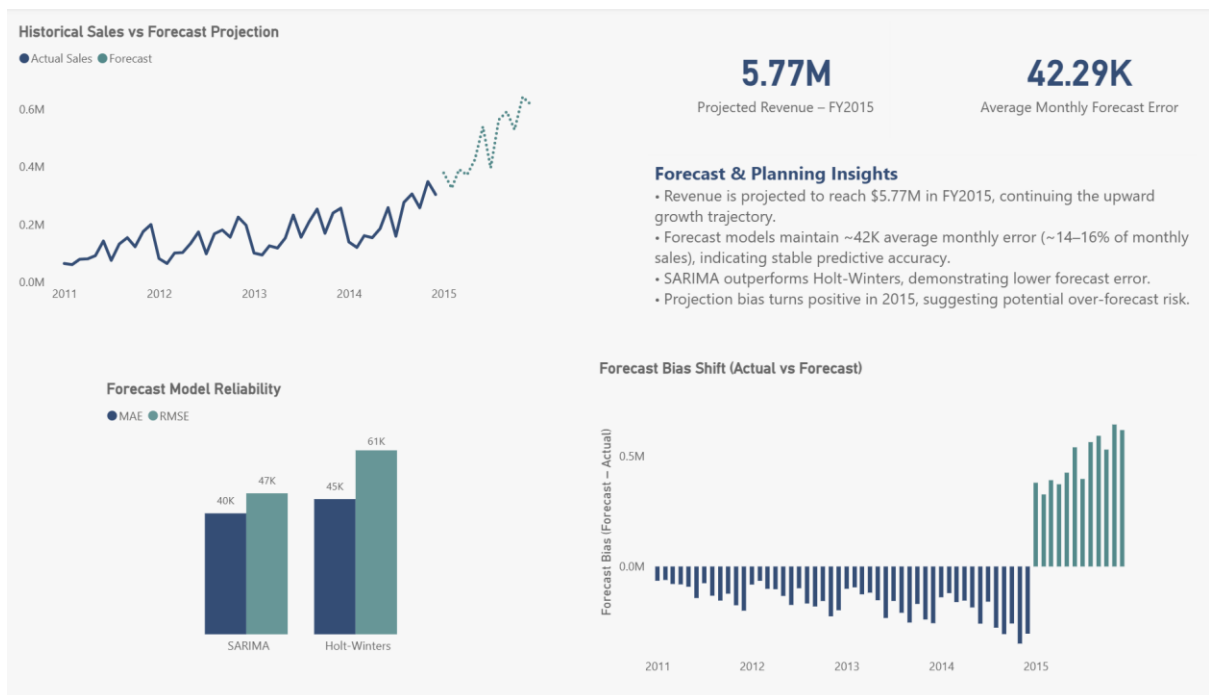
Customer segmentation analysis indicates that the **Consumer segment generates the largest share of revenue**, followed by Corporate and Home Office segments. Regional analysis highlights **APAC as a strong revenue contributor**.



Logistics & Operational Efficiency

Shipping analysis shows **Standard Class shipments dominate order volume**, while average delivery time is concentrated around **four days**, providing a benchmark for operational efficiency.

Forecasting Analysis



Forecasting Approach

To extend the analysis beyond historical performance, time-series forecasting models were developed using Python. Monthly revenue data aggregated through SQL queries was used as the input for predictive modeling.

Two forecasting models were applied:

SARIMA (Seasonal AutoRegressive Integrated Moving Average)

Captures seasonality and long-term trends in time-series sales data.

Holt-Winters Exponential Smoothing

Models trend and seasonal components using weighted smoothing techniques.

Model performance was evaluated using standard forecasting metrics including **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)**.

Forecast Results

The forecasting models project continued revenue growth based on historical sales patterns.

Key outcomes include:

- Projected revenue of approximately **\$5.77M for FY2015**
- Average monthly forecast error of around **\$42K**
- Overall model accuracy estimated within a **14–16% error range**

Among the evaluated models, **SARIMA demonstrated stronger predictive performance**, producing lower forecast error across validation periods.

Key Analytical Findings

Based on the full analysis across dashboards and forecasting models, several important insights emerge:

- **Revenue Growth:** Historical sales show a consistent upward trend with seasonal demand patterns.
- **Profitability Risk:** Profit margins decline significantly when discounts exceed approximately **40%**, indicating pricing sensitivity.
- **High-Performing Categories:** Products such as **Copiers and Phones** contribute disproportionately to overall profits.
- **Customer Segment Contribution:** The **Consumer segment generates the largest share of revenue**, highlighting its importance to overall sales performance.
- **Operational Efficiency:** Standard Class shipping dominates order volume, with delivery times concentrated around **four days**.

Strategic Implications

These insights highlight several potential areas for business focus:

- Optimizing discount strategies to protect profit margins
- Prioritizing high-performing product categories for revenue growth
- Strengthening engagement with high-value customer segments
- Using predictive forecasting to support financial planning and demand forecasting